

ALISON EMILIEN

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SUMMARY

Electrical Engineering and Computing Technology graduate focused on ASIC front-end design, functional verification, behavioral modeling, and open-source EDA flows. Hands-on experience building RTL modules, self-checking testbenches, Python automation, waveform-debug workflows, SystemC/C++ DSP models, and Sky130A transistor-to-layout projects. Strong fit for RF/Analog verification roles involving mixed-signal awareness, event-driven simulation, register/control logic, state machines, failed-test diagnosis, and AI-assisted verification methodology development.

PROJECTS

AI-Assisted RTL Analysis and OpenLane Verification Agent

Python, FastAPI, cocotb-style workflows, Icarus Verilog, OpenLane, OpenROAD

- Built a local ASIC/verification assistant for Verilog simulation, RTL debug, waveform discovery, and OpenLane/OpenROAD flow review.
- Integrated backend tooling with Icarus Verilog and cocotb-style simulation workflows to automate test execution and artifact collection.
- Designed workflow around failed-test diagnosis, structured log review, and AI-assisted explanations for RTL and physical-design feedback.

RF/DSP Receiver Modeling and Verification Project

SystemC, C++, Verilog, Python, fixed-point modeling, QPSK, FIR, GTKWave

- Developed a SystemC/C++ receiver-style project with QPSK sample generation, FIR filtering, fixed-point modeling, and demapping logic.
- Implemented matching Verilog RTL blocks for FIR filter and QPSK demapper to connect high-level behavioral models with hardware logic.
- Created Python scripts to generate input stimulus, compare expected-versus-actual outputs, and support repeatable self-checking verification.
- Produced waveform outputs for signal-path debug and validated datapath behavior through model-to-RTL comparison.

Custom D Flip-Flop and Open-Source ASIC Flow | Sky130A

Xschem, Ngspice, Magic VLSI, KLayout, Netgen, Sky130A

- Designed and verified a transistor-level master-slave D flip-flop using the Sky130A 130 nm CMOS process.
- Captured schematics in Xschem, performed transient simulations in Ngspice, and created physical layout using Magic VLSI and KLayout.
- Practiced DRC/LVS-style debug with Netgen and layout review to connect schematic intent, extracted views, and physical implementation constraints.
- Strengthened understanding of sequential logic, setup/hold behavior, analog/digital boundary effects, and ASIC implementation flow.

Cocotb Bus/Register Verification Environment

Verilog, cocotb, Python, Makefile, Icarus Verilog, GTKWave

- Built a cocotb-based verification project for bus/register logic using Python stimulus, assertions, reset tests, and waveform-driven debug.
- Verified register read/write behavior, control/status updates, reset handling, and edge-case sequences with self-checking tests.
- Automated simulation execution through Makefile-based workflows using Icarus Verilog and GTKWave/VCD output.

Standard Cell Characterization and Timing Analysis

CMOS inverter analysis, transient simulation, timing measurement, layout validation

- Conducted transistor-level transient analysis on CMOS inverter behavior and measured propagation-delay differences between rising and falling transitions.
- Quantified timing behavior, identified device asymmetry, and explored width-ratio optimization to improve switching balance.
- Validated physical integrity of CMOS logic layouts through DRC-focused layout review and timing-oriented analysis.

SKILLS

Programming Languages: Python, C, C++, MATLAB, Verilog, SystemVerilog fundamentals, Bash/Zsh

Verification: cocotb, UVM, self-checking testbenches, assertions, coverage planning concepts, regression-style debug, waveform analysis

ASIC/EDA Tools: OpenLane, OpenROAD, Icarus Verilog, GTKWave, Xschem, Ngspice, Magic VLSI, KLayout, Netgen, Sky130A PDK, Quartus II

Modeling/Signal Processing: SystemC, fixed-point modeling, FIR filters, QPSK demapping, MATLAB/Simulink, transient simulation, timing characterization

AI/ML: PyTorch, scikit-learn, supervised/unsupervised learning fundamentals, prompt-based AI workflows, AI-assisted EDA/debug tooling

WORK EXPERIENCE

University of Ottawa - Undergraduate Teaching Assistant

Ottawa, ON | May 2023 – Apr 2026

- Mentored students through C programming, digital systems, logic design, Quartus II/Altera workflows, and lab-style debugging.
- Reviewed student code and hardware-oriented assignments, helping identify logic errors, syntax issues, and design-intent mismatches.
- Explained complex computing and digital-logic concepts in concise, reproducible steps during lab evaluations.

University of Ottawa - Pension and Benefits Assistant

Ottawa, ON | May 2023 – Apr 2026

- Maintained high-accuracy records, structured logs, and process documentation across deadline-driven administrative workflows.
- Deconstructed complex data issues into traceable workstreams and documented bottlenecks for future process improvement.
- Built discipline in confidentiality, auditability, issue tracking, and detail-oriented review for high-reliability work.

EDUCATION

University of Ottawa — BAsC Electrical Engineering and BSc Computing Technology

Ottawa, ON | Sep 2020 – Apr 2026

Relevant Coursework: VLSI Design, Digital Systems, Computer Architecture, Electronics, Control Systems, Software Engineering